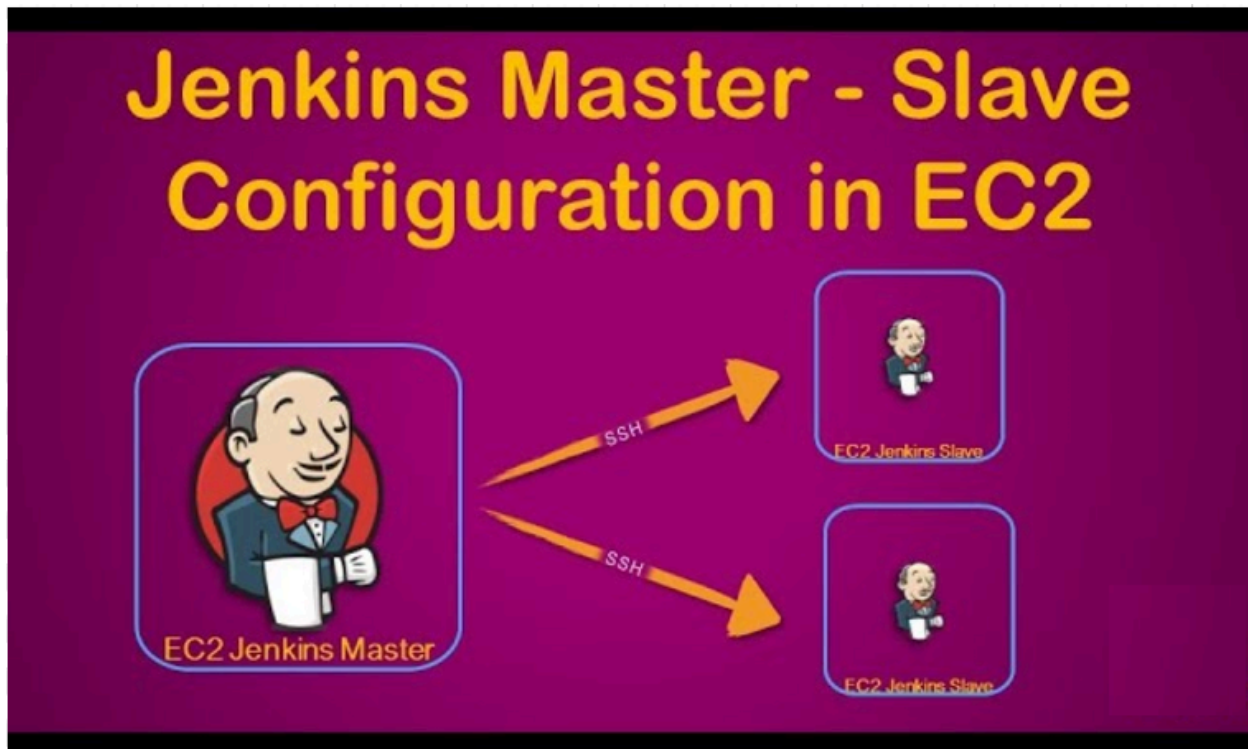
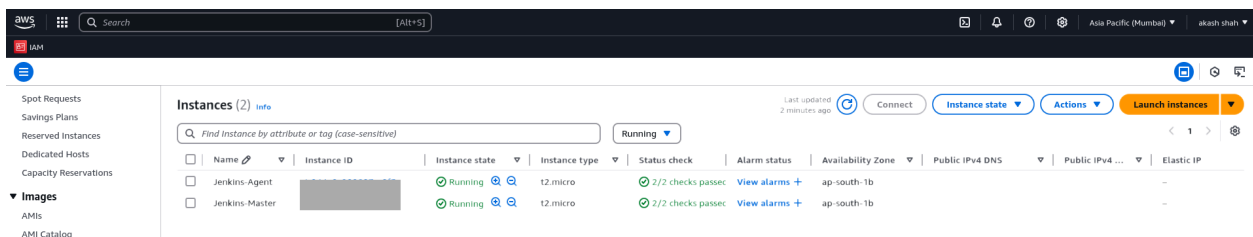


Jenkins on AWS EC2



Launch EC2 Instance on AWS

1. Launch an EC2 instance using your preferred OS (Ubuntu recommended).
2. Modify the EC2 security group to allow inbound traffic on port **8080** (for Jenkins Web Interface).



3. Take ssh access to the Jenkins-master instance.
`ssh -i /path/to/your-key.pem ubuntu@<ec2-public-ip>`

Install Java on the master and agent node (Required for Jenkins)

1. Update the system:
`sudo apt update; sudo apt upgrade`
2. Install Java (Jenkins requires Java to run):
`sudo apt install openjdk-17-jre`
3. Verify Java installation:
`java -version`

Install Jenkins on the Master Node

- ♦ Follow the Jenkins installation instructions for Ubuntu:

[Jenkins Installation](#)

- ♦ Add Jenkins repository key:

```
sudo wget -O /usr/share/keyrings/jenkins-keyring.asc \
https://pkg.jenkins.io/debian-stable/jenkins.io-2023.key
```

- ♦ Add the Jenkins repository:

```
echo "deb [signed-by=/usr/share/keyrings/jenkins-keyring.asc] \
https://pkg.jenkins.io/debian-stable binary/" | sudo tee \
/etc/apt/sources.list.d/jenkins.list > /dev/null
```

- ♦ Update and Install Jenkins:

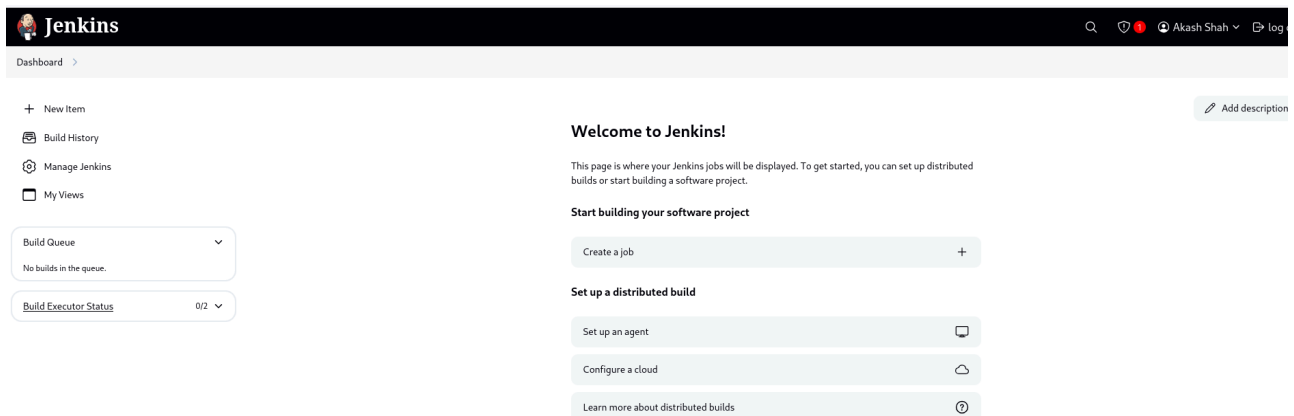
```
sudo apt-get update
sudo apt-get install jenkins
```

- ♦ Enable and Start Jenkins to start on boot:

```
sudo systemctl enable jenkins
sudo systemctl start jenkins
```

- ♦ Check the Jenkins service status:

```
systemctl status jenkins
```



Configure SSH on Jenkins Master

1. Edit SSH configuration:

```
sudo vi /etc/ssh/sshd_config
```

Enable `PubkeyAuthentication` and `AuthorizeKeysFile`.

2. Reload SSH service:

```
sudo service sshd reload
```

Configure Jenkins-Agent Node

Repeat the steps mentioned in the **Install Java** and **Configure SSH** sections for the Jenkins Agent instance:

- ❖ **Install Java** on the Agent Instance.

- ❖ Edit SSH configuration:

```
sudo vi /etc/ssh/sshd_config
```

Enable `PubkeyAuthentication` and `AuthorizeKeysFile`.

- ❖ Reload SSH service:

```
sudo service sshd reload
```

- ❖ Generate SSH key pair:

```
ssh-keygen OR ssh-keygen -t ed25519
```

```
ubuntu $ ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/root/.ssh/id_rsa):
/root/.ssh/id_rsa already exists.
Overwrite (y/n)? Y
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /root/.ssh/id_rsa
Your public key has been saved in /root/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:0WJ1Iy5NFeDmw+uqkabSCdBxN9cJRnUkSGXy5XdcWfc root@ubuntu
The key's randomart image is:
+---[RSA 3072]-----+
|      o**X=0.  *|
|    . . o.o@oB . o+|
|  . o . o= B . . E|
| . . . B . . .|
| . . . S +|
| . . . o|
|  o .+ .|
| . oo . .|
|  .. ..|
+-----[SHA256]-----+
ubuntu $ cd .ssh/
ubuntu $ ls
authorized_keys  id_rsa  id_rsa.pub  known_hosts
```

❖ Copy the public key to the Jenkins Agent:

```
cd /home/ubuntu/.ssh
```

```
cat id_rsa.pub # Copy the key to Jenkins Agent.
```

❖ Add the Jenkins-Agent server Public Key to the authorized_keys:

```
ssh-keygen
```

```
> cd /home/ubuntu/.ssh
```

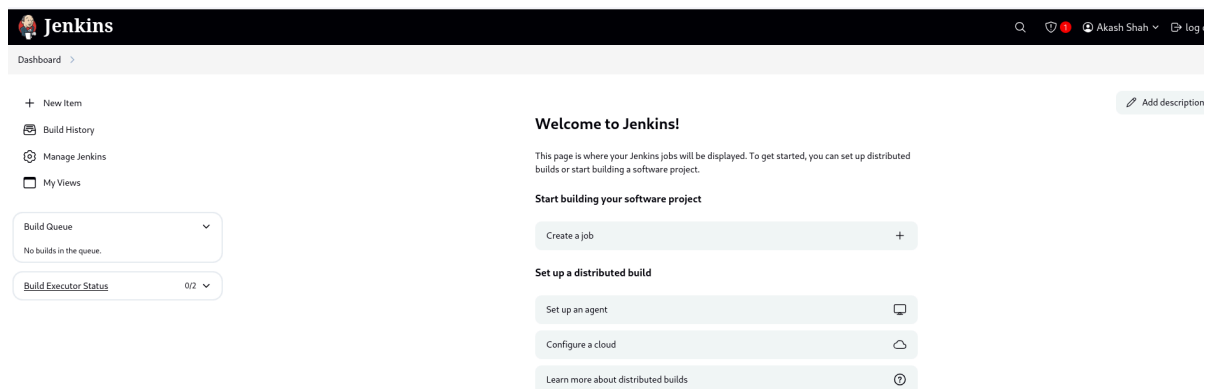
```
> cat id_ed25519.pub
```

```
> vi authorized_keys # Add the above ssh public key.
```

Configure Jenkins Agent on Jenkins Master

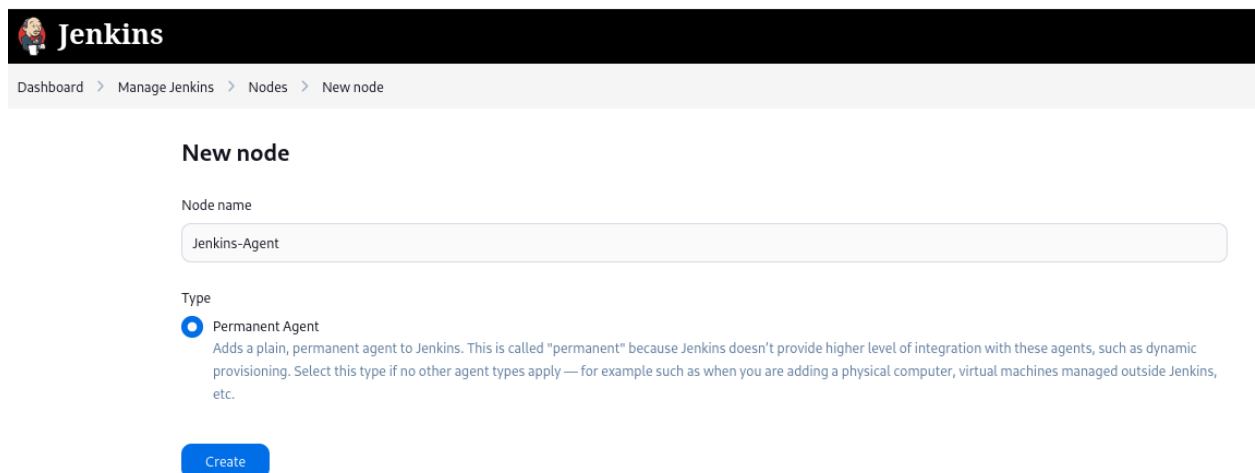
1. Access Jenkins Dashboard:

- Go to <http://<jenkins-master-ip>:8080> and log in.



2. Navigate to:

[Manage Jenkins](#) → [Manage Nodes and Clouds](#) → [New Node](#)



3. Create New Node:



The screenshot shows the Jenkins 'New Node' configuration page. The breadcrumb trail is 'Dashboard > Manage Jenkins > Nodes'. The form fields are as follows:

- Name:** Jenkins-Agent
- Description:** Jenkins-Agent
- Number of executors:** 4
- Remote root directory:** /home/ubuntu
- Labels:** Agent-Node
- Usage:** Use this node as much as possible
- Launch method:** Launch agents via SSH

- **Node Name:** Jenkins-Agent
- **Type:** Permanent Agent
- **Number of Executors:** 2 (or as needed)
- **Remote Root Directory:** /home/ubuntu
- **Labels:** Jenkins-Agent
- **Usage:** Use this node as much as possible

4. Launch Method:

- **Launch agents via SSH**

5. Configure SSH Connection:



The screenshot shows the 'Configure SSH Connection' page for the Jenkins node. The breadcrumb trail is 'Dashboard > Manage Jenkins > Nodes'. The form fields are as follows:

- Usage:** Use this node as much as possible
- Launch method:** Launch agents via SSH
- Host:** 172.30.1.2
- Credentials:** ubuntu (Jenkins-Agent node key)
- Host Key Verification Strategy:** Known hosts file Verification Strategy
- Availability:** Keep this agent online as much as possible
- Node Properties:**
 - ☐ Disable deferred wipeout on this node
 - ☐ Disk Space Monitoring Thresholds
 - ☐ Environment variables
 - ☐ Tool Locations

A 'Save' button is located at the bottom of the form.

- **Host:** Enter the private IP of the Jenkins Agent EC2 instance.

- **Add Credentials:**

New credentials

Kind
SSH Username with private key

Scope
Global (Jenkins, nodes, items, all child items, etc)

ID
Jenkins-Agent

Description
Jenkins-Agent node key

Username
ubuntu

☐ Treat username as secret

Private Key
☒ Enter directly

Key
-----BEGIN OPENSSH PRIVATE KEY-----
b3B1bnlnZaC1rZktdjEAAAABG5vbmUAAAABbm9uZQAAAAAAAAABAAAwAAATzc2g1ZW
DvNTUxODAAAC1Rk5uJ0R0RnT4+WbTw1KIDd67NSeuerC7R0dEF7R7P+hoAAAAJID0089Kwz1D

Passphrase

Create

- **Kind:** SSH Username and Private Key
- **Username:** `ubuntu`
- **Private Key:** Paste the Jenkins Agent's private key.

- **Host Key Verification Strategy:** `Non-verifying Verification Strategy`

6. Save the Node Configuration.

Check Added Node and Build Job

1. Monitor the Jenkins Master and Agent logs to ensure they are communicating properly.

Jenkins Dashboard - Nodes

S	Name	Architecture	Clock Difference	Free Disk Space	Free Swap Space	Free Temp Space	Response Time
	Built-In Node	Linux (amd64)	In sync	3.73 GiB	0 B	3.73 GiB	0ms
	Jenkins-Agent	Linux (amd64)	In sync	10.99 GiB	0 B	10.99 GiB	84ms
Data obtained				23 sec	23 sec	23 sec	23 sec

Icon: S M L

- On the Jenkins Dashboard, **Build a Job** and check if the job runs on the Jenkins Agent node.

The screenshot displays the Jenkins Dashboard interface. At the top, the Jenkins logo and user information (Akash Shah) are visible. The left sidebar contains navigation links: New Item, Build History, Project Relationship, Check File Fingerprint, Manage Jenkins, and My Views. Below these are two expandable sections: 'Build Queue' (showing 'No builds in the queue.') and 'Build Executor Status' (showing 'Jenkins-Agent' with '0/2' capacity). The main content area features a table of build history for the 'test-project' job. The table has columns for status (S), warnings (W), name, last success, last failure, and last duration. A single build is listed with a success status, a warning icon, a duration of 12 seconds, and a last failure of N/A. A green play button icon is visible in the bottom right corner of the table.

S	W	Name	Last Success	Last Failure	Last Duration
✓	⚠	test-project	12 sec #1	N/A	4.3 sec